

A Practical Guide to Retrieval augmented generation

TOPIC -- RETRIEVAL AUGMENTED GENERATION

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Hybrid search Sometimes vector database searches can miss key facts needed to answer a user's question. One way to mitigate this is to do a traditional text search, add those results to the text chunks linked to the retrieved vectors from the vector search, and feed the combined hybrid text into the language model for generation. The adoption of RAG in consumer-facing web search products has given rise to new content optimization disciplines, as practitioners have noted that content retrievability in RAG systems depends on factors like semantic structure, passage-level authority signals, and entity clarity rather than traditional search ranking signals such as backlinks.

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Retrieval-augmented generation (RAG) is a technique that enables large language models (LLMs) to retrieve and incorporate new information from external data sources. With RAG, LLMs first refer to a specified set of documents, then respond to user queries. These documents supplement information from the LLM's pre-existing training data. This allows LLMs to use domain-specific and/or updated information that is not available in the training data. For example, this helps LLM-based chatbots access internal company data or generate responses based on authoritative sources. RAG improves LLMs by incorporating information retrieval before generating responses. Unlike LLMs that rely on static training data, RAG pulls relevant text from databases, uploaded documents, or web sources. According to Ars Technica, "RAG is a way of improving LLM performance, in essence by blending the LLM process with a web search or other document look-up process to help LLMs stick to the facts." This method helps reduce AI hallucinations, which have caused chatbots to describe policies that don't exist, or recommend nonexistent legal cases to lawyers that are looking for citations to support their arguments. RAG also reduces the need to retrain LLMs with new data, saving on computational and financial costs. Beyond efficiency gains, RAG also allows LLMs to include sources in their responses, so users can verify the cited sources. This provides greater transparency, as users can cross-check retrieved content to ensure accuracy and relevance. The term retrieval-augmented generation (RAG) was introduced in a 2020 paper that described combining a parametric language model with a non-parametric external memory accessed through retrieval at

inference time.

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Encoder These methods focus on the encoding of text as either dense or sparse vectors. Sparse vectors, which encode the identity of a word, are typically dictionary-length and contain mostly zeros. Dense vectors, which encode meaning, are more compact and contain fewer zeros. Various enhancements can improve the way similarities are calculated in the vector stores (databases).

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Typically, the data to be referenced is converted into LLM embeddings, numerical representations in the form of a large vector space. RAG can be used on unstructured (usually text), semi-structured, or structured data (for example knowledge graphs). These embeddings are then stored in a vector database to allow for document retrieval. Given a user query, a document retriever is first called to select the most relevant documents that will be used to augment the query. This comparison can be done using a variety of methods, which depend in part on the type of indexing used. The model feeds this relevant retrieved information into the LLM via prompt engineering of the user's original query. Newer implementations (as of 2023) can also incorporate specific augmentation modules with abilities such as expanding queries into multiple domains and using memory and self-improvement to learn from previous retrievals. Finally, the LLM can generate output based on both the query and the retrieved documents. Some models incorporate extra steps to improve output, such as the re-ranking of retrieved information, context selection, and fine-tuning.

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Performance improves by optimizing how vector similarities are calculated. Dot products enhance similarity scoring, while approximate nearest neighbor (ANN) searches improve retrieval efficiency over K-nearest neighbors (KNN) searches. Accuracy may be improved with Late Interactions, which allow the system to compare words more precisely after retrieval. This helps refine document ranking and improve search relevance. Hybrid vector approaches may be used to combine dense vector representations with sparse one-hot vectors, taking advantage of the computational efficiency of sparse dot products over dense vector operations. Other retrieval techniques focus on improving accuracy by refining how documents are selected. Some retrieval methods combine sparse representations, such as SPLADE, with query expansion strategies to improve search accuracy and recall.

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Process Retrieval-augmented generation (RAG) enhances large language models (LLMs) by incorporating an information-retrieval mechanism that allows models to access and utilize additional data beyond their original training set. Ars Technica notes that "when new information becomes available, rather than having to retrain the model, all that's needed is to augment the model's external knowledge base with the updated information" ("augmentation"). IBM states that "in the generative phase, the LLM draws from the augmented prompt and its internal representation of its training data to synthesize" an answer.

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Pre-training the retriever using the Inverse Cloze Task (ICT), a technique that helps the model learn retrieval patterns by predicting masked text within documents. Supervised retriever optimization aligns retrieval probabilities with the generator model's likelihood distribution. This involves retrieving the top-k vectors for a given prompt, scoring the generated response's perplexity, and minimizing KL divergence between the retriever's selections and the model's likelihoods to refine retrieval. Reranking techniques can refine retriever performance by prioritizing the most relevant retrieved documents during training.

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Challenges RAG does not prevent hallucinations in LLMs. According to Ars Technica, "It is not a direct solution because the LLM can still hallucinate around the source material in its response." While RAG improves the accuracy of large language models (LLMs), it does not eliminate all challenges. One limitation is that while RAG reduces the need for frequent model retraining, it does not remove it entirely. Additionally, LLMs may struggle to recognize when they lack sufficient information to provide a reliable response. Without specific training, models may generate answers even when they should indicate uncertainty. According to IBM, this issue can arise when the model lacks the ability to assess its own knowledge limitations.

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Evaluation and benchmarks RAG systems are commonly evaluated using benchmarks designed to test retrievability, retrieval accuracy and generative quality. Popular evaluation resources include BEIR, a heterogeneous benchmark for zero-shot information retrieval across multiple datasets, and Natural Questions, a large-scale question answering dataset released by Google Research.